

SIDDHARTH DUDUGU

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PROFESSIONAL SUMMARY

Results-driven Data & Business Analyst with 3+ years of experience in data analysis, machine learning, and business intelligence. **Microsoft Certified Power BI Data Analyst (PL-300)**, **Tableau Certified Data Analyst** and **AWS Certified Data Engineer Associate** with proven expertise in predictive modelling, time-series analysis, anomaly detection, and ensemble modelling to enhance forecasting accuracy and optimize decision-making. Skilled in trend forecasting, performance analysis, customer segmentation, and KPI identification, delivering actionable insights that drive revenue optimization. Adept at evaluating price distribution, elasticity trends, and simulating pricing strategies to maximize ROI. Experienced in ETL, database management, and advanced analytics, leveraging Python, R, SQL, Excel, Tableau, Power BI, Snowflake, Alteryx, Databricks, along with cloud platforms like Azure, AWS, and GCP. Strong communicator, adept at translating complex data into strategic insights for cross-functional teams and stakeholders, influencing business outcomes and marketing tactics. Passionate about uncovering opportunities hidden within data to drive measurable business impact.

EDUCATION

Drexel University - Philadelphia, PA, USA
Master of Science in Data Science

Sep' 2022 – Jun' 2024
GPA: 3.91/4

National Institute of Technology (NIT) - Warangal, India
Post-Graduate Diploma in Artificial Intelligence (AI) and Machine Learning (ML)

Sep' 2019 – Jun' 2021
GPA: 3.57/4

University of Mumbai, India
Bachelor of Engineering in Computer Science

Aug' 2015 – Jun' 2019
GPA: 3.71/4

PROFESSIONAL EXPERIENCE

Data Research Analyst
Map Collective Inc.

Mar' 2024 – Dec' 2024
Washington, DC, USA

- Analyzed market trends, conversion rates, and revenue patterns across 50+ supply chains using SQL, Python (Pandas, NumPy), and Power BI, driving 15% improvement in data-backed decision-making for sustainable supply chain solutions.
- Maintained and updated 1M+ data records using advanced SQL, improving data integrity by 85%+ for tracking sustainability metrics and ensuring compliance with Europe Deforestation Regulation (EUDR) and biodiversity requirements.
- Developed predictive models and built machine learning models in Python (scikit-learn, TensorFlow) to forecast supply chain risks and opportunities, leading to 20% more accurate strategic planning and operational efficiency.
- Provided actionable insights by designing real-time Power BI dashboards and visualizations, translating complex data into 15+ dynamic reports, enabling cross-functional teams to make sustainability-driven decisions 30% faster.

Data Analyst Intern
Data Glacier

Oct' 2023 – Dec' 2023

- Enhanced Go-to-Market (G2M) strategy for the cab industry, increasing actionable insights by 70% through exploratory data analysis (EDA) using Python and Tableau, leading to data-driven investment recommendations.
- Improved retail forecasting accuracy by 20% by leading projects that utilized advanced machine learning algorithms, multivariate forecasting models, and time series analysis for demand prediction.
- Implemented efficient data collection, cleaning, and pre-processing while building optimized ETL pipelines using Snowflake, reducing duplicates by 50% and enhancing data modelling accuracy with Excel, Python, and Tableau.
- Created insightful dashboards and reports using Tableau and applied version control using Git, adhering to Agile/Scrum methodologies using JIRA.

Business Data Analyst
Innوبراند Media and Communications

Aug' 2019 – Jul' 2022
Mumbai, Maharashtra, India

- Extracted data from multiple sources using Python, transformed, and analyzed data using Power BI, developing and maintaining ETL pipelines and automation scripts to enhance data processing efficiency and scalability.
- Increased television, digital media ratings by 15% through strategic content analysis and targeted program scheduling based on audience demographics, viewership trends, and competitor analysis, leveraging advanced SQL, Power BI and Excel for data-driven insights.
- Boosted TV and digital media ad ROI by 20% through market research, performance analysis, and time-series modeling, optimizing time slot allocations for peak audience engagement.
- Enhanced advertising campaign effectiveness by 40% by conducting exploratory data analysis (EDA) using Power BI dashboards, identifying key performance indicators (KPIs), and detecting anomalies to drive data-informed decisions.

SKILLS

Languages: Python, R, SQL, Java, C, C++, HTML, CSS, DAX

Frameworks & Tools: Apache Spark, Hadoop, Kafka, TensorFlow, NumPy, Pandas, NLTK, Keras, Sci-Py, Databricks, Matplotlib, Flask, BeautifulSoup, Git, Power BI, Microsoft 365, Tableau, MS Office, QuickSight, MS Excel (Pivot Tables, Macros, VBA, VLOOKUP),

Databases: Amazon Redshift, Microsoft SQL Server (SSMS), MySQL, PostgreSQL, MongoDB, Oracle Database, Snowflake

Cloud: AWS (EC2, S3, DynamoDB, Redshift), Microsoft Azure, GCP

Methodologies: Agile, Scrum, Kanban, Lean Six Sigma (Green Belt), JIRA, Confluence, Visio, Project Libre, Trello, Salesforce, MS Access

ACADEMIC PROJECTS

- **Stock Market Portfolio Optimization**- Analyzed historical market trends, calculating expected returns, volatilities, and stock correlations to inform portfolio management strategies. Implemented Modern Portfolio Theory (MPT) and Monte Carlo simulations to dynamically adjust asset allocations, resulting in a 15% increase in risk-adjusted returns compared to static methods.
- **Customer Analytics** - Performed customer segmentation and personality analysis on marketing campaign data, identifying key customer segments through clustering and understanding how customers interact with a business, product, or service. Utilized Python libraries such as Pandas & Scikit-learn for data preprocessing, clustering, and Power BI dashboards and reports to drive targeted marketing strategies.
- **Dynamic Price Optimization** - Developed a dynamic pricing model by analyzing price elasticity, sales performance, and competitor pricing across multiple categories, improving revenue potential through strategic price adjustments based on customer demand. Implemented advanced BI and data visualization using Tableau to compare price distribution, evaluate elasticity trends, and simulate pricing strategies.
- **U.S. Electric Vehicle Market Growth and Trend Analysis** - Analyzed the market data using Python – Pandas and NumPy, focusing on historical registration data to assess market growth, EV adoption trends, and geographical distribution. Applied statistical modeling techniques to forecast future EV registrations and identified key factors driving market expansion. Delivered insights on EV model popularity, electric range advancements, and strategic recommendations in production, infrastructure, and policy planning.
- **Crypto-currency Price Prediction** - Analyzed crypto prices, utilizing feature engineering, statistical methods, and EDA, focusing on the 40th to 60th data points. Utilized Linear Regression and LSTM algorithm (Deep Learning) approaches, achieving 93% precision, forecasting a decline in the value.
- **Cab Booking System** - Performed time-series analysis by integrating historical usage patterns & open data sources, time-forecasting the cab booking. Executed end-to-end data processing, including grid search, ensemble modeling techniques, achieving 93% accuracy through regression analysis, while incorporating feature selection, engineering and k-fold cross-validation to optimize model performance.
- **Time-Series Forecasting** - Implemented ARIMA model for time series forecasting, utilizing historical financial data using Python. Designed Power BI dashboard, applying conditional formatting for highlighting outliers and deviations, facilitating trend comparison, and enabling informed decision-making and implementing year-over-year (YoY) performance metrics calculations using DAX expressions.

CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate (PL-300)
- Tableau Certified Business Intelligence Analyst
- AWS Certified: Data Engineer Associate
- Boston Consulting Group (BCG) Data Science certificate
- Accenture - North America Data Analytics certificate
- PMI Six Sigma Foundations
- IIBA Six Sigma Green Belt
- PWC Power BI Analyst